

DESIGN TRANSFER PLAN

CSI-100 Cell Sorting Instrument (IVD)

Document number	DTP-CSI-001
Revision	B
Document type	Plan
Status	Released
Effective date	September 2026
Author	Atif Lodhi, Systems Engineering
Product	CSI-100 Cell Sorting Instrument, all configurations

Purpose

This plan defines the transfer of the CSI-100 design from R&D to manufacturing, including the product technical description, configurations, factory specifications, Device Master Record readiness, process validation, suppliers, training, and transfer acceptance criteria.

Document Control

Rev	Date	Description of change	Author
A	Sep 2026	Initial release after Design Transfer Readiness Review #1.	A. Lodhi
B	Sep 2026	Added Section 3 Product Technical Description (intended use, operating principle, laser, optics, fluidics and sort specifications, configuration matrix, options, consumables, service), Section 7 Factory Build, Alignment and Test Specifications, Section 10 Manufacturing Equipment and Tooling, an approved vendor list in Section 11, source document citations for all specifications, and Section 16 Reference Documents.	A. Lodhi

Approvals

Function	Role	Signature / Date
Systems Engineering	Transfer Lead (author)	
R&D Engineering	Design Owner	
Manufacturing Engineering	Process Owner (receiving)	
Quality Engineering	Design / Process Quality	
Supply Chain	Supplier Quality & Procurement	
Regulatory Affairs	Submission & Labeling	
Service & Support	Field Readiness	

Contents

1. Purpose and Scope
 2. Applicable Standards
 3. Product Technical Description
 4. Transfer Organization and Responsibilities
 5. Transfer Strategy and Phases
 6. Device Master Record (DMR) Readiness
 7. Factory Build, Alignment and Test Specifications
 8. Manufacturing Process Validation
 9. Test Method and Calibration Transfer
 10. Manufacturing Equipment, Tooling and Facilities
 11. Supplier Readiness and Approved Vendor List
 12. Training and Qualification of Personnel
 13. Risk Management in Transfer
 14. Transfer Acceptance Criteria
 15. Post-Transfer Monitoring and Handoff Closure
 16. Reference Documents
- Appendix A. Terms and Abbreviations

1. Purpose and Scope

This plan defines how the CSI-100 Cell Sorting Instrument moves from R&D design control into controlled, repeatable production. It gives manufacturing a complete technical picture of the product (what it does, how it works, every configuration, the hardware and vendors involved and the specifications each unit must meet) and defines who owns each transfer element, how readiness is demonstrated with objective evidence, and the criteria for manufacturing to take ownership.

Every specification in this plan cites the controlled R&D document it comes from, by document number and section. The revision of each cited document is listed in Section 16, Reference Documents. If a value in this plan differs from the cited document, the cited document governs and this plan is corrected through document change control. Factory limits come from the Factory Alignment and Test Specification (FTS-CSI-001). They are set tighter than the system requirements to allow for measurement uncertainty, shipping and field drift, and are confirmed by pilot build capability data before release.

In scope

- Final assembly, subsystem build, optical alignment and calibration of the sorter at the production site.
- Build and test of the fluidics supply module, including its umbilical to the sorter.
- All released factory laser configurations (2- to 6-laser builds) and field-upgrade kits for lasers, detector arrays and additional PMTs.
- Instrument control software and firmware installation and configuration at manufacturing.
- Manufacturing test, final acceptance test (FAT), burn-in, packaging, labeling and shipping of both units.
- Options built on the same line: automated plate deposition unit, aerosol evacuation module, collection temperature control.
- Accessory kit and ship-with consumables, and the documentation service needs to install, verify and maintain the instrument.

Out of scope

- IVD reagents, assay kits and setup/QC beads as products (transferred under their own plans; this plan covers their use as test materials).
- Design changes after transfer, which are managed through the Engineering Change process.
- Commercial launch activities other than those gating product release.

Most production nonconformances on complex instruments trace back to knowledge that lived only in R&D: an alignment trick, an undocumented torque, a test limit set by feel. This plan makes that knowledge explicit, verifiable and owned before the handoff, not after the first field failure.

2. Applicable Standards

The following external standards and regulations apply to this transfer. The CSI-100 program documents cited throughout this plan, with their revisions, are listed in Section 16, Reference Documents.

Reference	Title / Application
21 CFR Part 820 (QMSR)	Quality Management System Regulation; incorporates ISO 13485:2016 by reference
ISO 13485:2016 §7.3.8	Design and development transfer

Reference	Title / Application
ISO 14971:2019	Application of risk management to medical devices
IEC 61010-1 / -2-101	Safety of laboratory equipment; particular requirements for IVD equipment
IEC 61326-1 / -2-6	EMC requirements; particular requirements for IVD medical equipment
IEC 60825-1	Safety of laser products (Class 1 product with embedded Class 3B sources)
IEC 62304 / IEC 82304-1	Software life cycle; health software product safety
ISO 14644-1	Cleanroom classification (flow cell and optics assembly area)
ANSI/ESD S20.20	ESD control program for electronics and detector handling
ISTA 2A / 3E	Transport packaging performance testing
EU IVDR 2017/746	Where the instrument is placed on the EU market
GHTF/SG3/N99-10	Process validation guidance (IQ / OQ / PQ)

3. Product Technical Description

This section is written for the people who will build, test and release the CSI-100. It explains what the product is for, how it works, and what hardware and performance each unit must deliver. The operator's guide explains how to use the instrument; this section explains what manufacturing must reproduce, and why each feature matters.

3.1 Intended use

The CSI-100 Cell Sorting Instrument is an automated, multi-laser flow cytometer and droplet cell sorter intended for the identification, enumeration and physical separation of cell populations in human specimens (such as peripheral blood, bone marrow and cellular therapy products) stained with fluorochrome-labeled reagents. It is intended for in vitro diagnostic use by trained laboratory professionals in clinical laboratories, when used with IVD reagents and assays validated for the instrument. Sorted cells are recovered for further laboratory analysis.

The intended users are trained clinical laboratory scientists. Installation and repair are performed by trained service engineers only. The instrument is used on a clinical laboratory bench and has no special room requirements; the user facility is responsible for a biosafety assessment before sorting potentially infectious specimens.

The CSI-100 is a Class II IVD instrument in the US (510(k)). Classification under EU IVDR is set by Regulatory Affairs according to the assays claimed.

The intended use determines which characteristics are critical for manufacturing. Fluorescence resolution, sort purity and aerosol containment affect patient results and operator safety, so they are controlled on every unit rather than by sampling.

3.2 Principle of operation

Fluidics. Pressurized sheath fluid from the fluidics supply module surrounds the sample in a quartz cuvette flow cell, hydrodynamically focusing cells into a single-file core stream. The sheath-to-sample pressure difference sets the core diameter and sample rate (about 10–80 $\mu\text{L}/\text{min}$). In the cuvette, cells travel slowly (about 6 m/s with the 70 μm setup), which lengthens laser exposure; the stream then accelerates to about 30 m/s at the nozzle exit.

Excitation and detection. Up to four laser beams, delivered by optical fiber and focused as spatially separated spots, cross the core stream. Each cell scatters light (forward scatter for relative size, side scatter for internal complexity) and emits fluorescence from bound dyes. A high-NA objective gel-coupled to the cuvette collects the light from each spot into its own collection fiber, which carries it to a detector array. Each array uses a cascade of longpass dichroic mirrors and bandpass filters to route wavelength bands to individual photomultiplier tubes (PMTs).

Signal processing. Each detector produces a pulse a few microseconds long. The electronics digitize the pulses and compute height, area and width, apply inter-laser delay so that signals from different beam spots line up for the same cell, and compare each event with user-defined sort gates in real time.

Sorting. A piezoelectric drop drive vibrates the stream so that it breaks into uniform droplets a few millimeters below the nozzle. When a target cell reaches the breakoff point (the drop delay), the stream is charged through a charging wire so that the droplet leaving at that moment carries a set positive or negative charge. The charged droplet passes between two high-voltage deflection plates and is steered into one of up to four collection streams; uncharged droplets go to a central waste aspirator. A breakoff camera and closed-loop drop-drive control hold the breakoff point steady, and a side-stream camera with a 660 nm diode laser is used to set and verify drop delay.

3.3 System architecture

Transfer is organized around the ten-subsystem architecture used throughout design, verification and risk management. Every deliverable in this plan traces to one of these subsystems, which keeps the DMR, the pFMEA and the manufacturing routings aligned with the design.

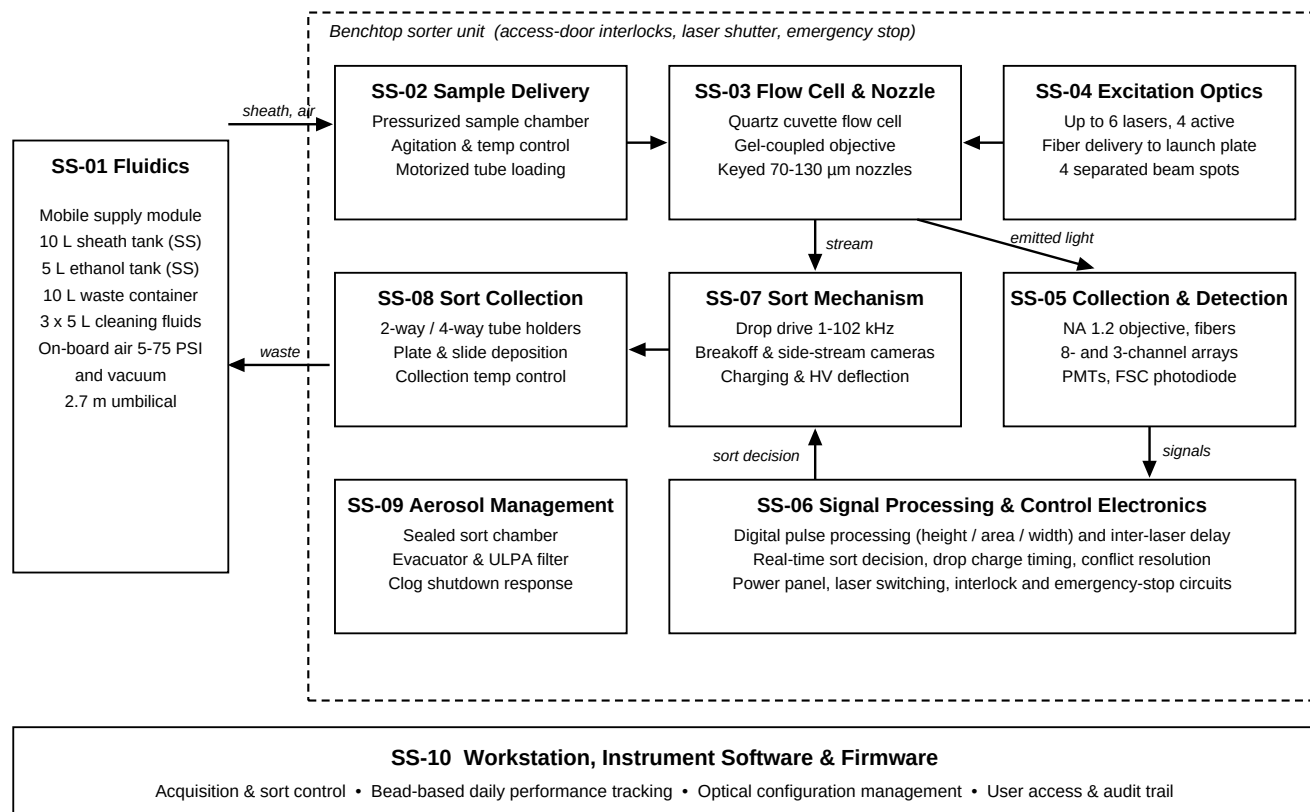


Figure 1. CSI-100 functional architecture. Sample and sheath meet in the flow cell, where the laser beams interrogate the stream; the detected signals drive the sort decision that charges and deflects droplets into collection.

ID	Subsystem	Build location	Complexity	Key transfer concern
SS-01	Fluidics supply module	Contract mfr	High	Air/vacuum regulation, tank pressure integrity, level sensing, umbilical keying
SS-02	Sample delivery	In-house	Medium	Chamber pressure seal, tube-loading motion safety, temperature control, carryover flush
SS-03	Flow cell & nozzle	In-house (cleanroom)	Very high	Optical gel coupling to objective, nozzle keying repeatability, cleanliness
SS-04	Excitation optics	Laser vendors + in-house	Very high	Fiber launch and beam-spot alignment retention through shipping and thermal cycling
SS-05	Collection & detection	In-house	High	Collection-fiber position, mirror/filter placement per configuration, PMT gain matching
SS-06	Signal processing & control	PCBA vendor + in-house	Medium	Board test coverage, firmware load control, interlock and e-stop verification
SS-07	Sort mechanism	In-house	High	HV deflection safety, camera calibration, sort block pivot setting
SS-08	Sort collection	In-house	Medium	Collection holder alignment, plate deposition accuracy
SS-09	Aerosol management	Final assy + vendor	High	Chamber sealing, evacuation flow, ULPA filter integrity

ID	Subsystem	Build location	Complexity	Key transfer concern
SS-10	Workstation & software	Software release	Medium	Version control, configuration file matched to installed optics

3.4 Physical, electrical and environmental specifications

Parameter	Sorter (benchtop unit)	Fluidics supply module	Source document
Dimensions (H × W × D)	71 × 122 × 71 cm (28 × 48 × 28 in)	66 × 81 × 66 cm (26 × 32 × 26 in)	SRS-CSI-001 §3.1; DWG-CSI-0001
Weight	≈ 181 kg (400 lb)	81.7 kg (180 lb) without tanks	SRS-CSI-001 §3.1
Optional bench (H × W × D)	85.1 × 266.7 × 134.6 cm	—	SPG-CSI-001 §2.1
Power	100 / 115 / 230 VAC, 50/60 Hz; 1,500 W maximum; single 20 A outlet	Powered from sorter only (voltage selector set at factory)	SRS-CSI-001 §3.2; ICD-CSI-010 §4
Operating environment	15–30 °C; 5–80% RH non-condensing	Same as sorter	SRS-CSI-001 §3.3
Storage environment	1–40 °C	Same as sorter	SRS-CSI-001 §3.3
Acoustic noise	≤ 62 dBA	Included in system measurement	SRS-CSI-001 §3.4
Facilities	No special room requirements	No external air or vacuum needed; optional house air 80–100 PSI, dry and oil-free	SPG-CSI-001 §3
Interconnect	Umbilical: power, fluid lines, air line, serial communication; up to 2.7 m (9 ft)	Connection panel for aerosol and temperature options	ICD-CSI-010 §3
Workstation link	Ethernet to instrument PC (fixed IP)	—	ICD-CSI-020 §2

3.5 Excitation optics and lasers

The system supports up to six installed lasers with up to four active at one time. The 488 nm and 633 nm lasers are standard; all others are options. Laser outputs are fiber-coupled to fiber heads on a common excitation launch plate. Beam-shaping prisms and an achromatic focusing lens place each laser on its own spot on the core stream. Because both the optical path and the core stream are fixed, alignment is set at the factory and is not adjusted by the user. The 375 nm laser is the exception in mounting: it is not on the launch plate but is mounted next to the flow cell.

λ (nm)	Color	Laser type	Beam shape	Min. power (at head)	Status	Beam spot
488	Blue	Solid-state, air-cooled	Elliptical	> 20 mW	Standard	Spot shared with 445
633	Red	Helium-neon (gas)	Elliptical	> 18 mW	Standard	Own spot
405	Violet	Solid-state	Elliptical	> 50 mW	Optional	Spot shared with 375
561	Yellow-green	Solid-state	Elliptical	> 50 mW	Optional	Fourth spot
445	Blue-violet	Solid-state	Elliptical	> 20 mW	Optional	488 path and PMTs
375	Near-UV	Solid-state, fixed mount	Top hat	> 7 mW	Optional	405 path and PMTs
660	Red (diode)	Diode, drop-delay system	—	—	Standard	Below deflection point

Source: SSD-CSI-400 §3 (laser specifications); CFG-CSI-001 §2 (laser options by build); SSD-CSI-700 §3.4 (660 nm drop-delay diode).

Excitation parameter	Specification	Source document
Beam spot size at interrogation point	Height $9 \pm 3 \mu\text{m}$ × width $65 \pm 7 \mu\text{m}$ (elliptical lasers)	SSD-CSI-400 §4.1
Beam spot arrangement	Up to 4 spatially separated spots; in 3-laser builds the stream meets red, then blue, then violet	SSD-CSI-400 §4.2
Inter-laser delay (70 μm setup, relative to blue)	Red -78, violet +39, yellow-green -39 (defaults; factory-verified)	SSD-CSI-600 §5.3
Laser switching	Rotary switches per laser (or laser pair) on the power panel; shared-spot pairs are mutually exclusive	SSD-CSI-600 §3.2
Warm-up	30 minutes after laser power-on before QC or acquisition	SSD-CSI-400 §4.5
Laser safety	Class 1 laser product with embedded Class 3B sources; flow cell door shutter interlock blocks beams when opened	SRS-CSI-001 §5.2; SVR-CSI-001
Cooling	Air-cooled; no special power or cooling needs for any laser	SSD-CSI-400 §4.6

3.6 Collection optics and detection

Light from each beam spot is collected by a fluorescence objective lens that is optically coupled to the quartz cuvette with refractive-index-matching gel (numerical aperture 1.2). The objective images each spot onto its own collection fiber; the fixed fiber apertures are spaced 200 μm apart to match the beam-spot separation. Each fiber feeds a detector array. The arrays pass the longest wavelengths to the first PMT and reflect shorter wavelengths onward through a cascade of longpass dichroic mirrors. Because reflection is more efficient than transmission, this layout maximizes multicolor sensitivity. A bandpass filter in front of every PMT selects the final band.

Detector element	Description	Source document
8-channel detector array	Cascade array holding up to 8 PMTs (positions A–H); used for the laser with the most colors	SSD-CSI-500 §3.1; DWG-CSI-5100
3-channel detector array	Cascade array holding up to 3 PMTs (positions A–C)	SSD-CSI-500 §3.2; DWG-CSI-5200
Maximum arrays	Two 3-channel arrays and two 8-channel arrays (second 8-channel array mounted behind the front arrays)	CFG-CSI-001 §2
Forward scatter (FSC)	Photodiode outside the arrays, 488/10 bandpass; removable ND filter (ND 1.0 installed; 0.5, 1.5, 2.0 supplied)	SSD-CSI-500 §3.4
Side scatter (SSC)	PMT with 488/10 bandpass (445/15 when the 445 nm laser is active)	SSD-CSI-500 §3.5
Filters	Longpass dichroic mirrors (angle-specific coatings) steer light; bandpass or discriminating filters select bands and block laser light	SSD-CSI-500 §4
Signal range	18-bit linear digital scale (2.6 to 262,144); log display over four or five decades	SSD-CSI-600 §4.2

3.7 Configuration matrix

Each laser combination is a separate controlled top-level assembly with its own BOM, array layout, mirror and filter map, software configuration file and FAT limits. The table below lists where the 8-channel arrays go for every released build. The 3-channel arrays serve the remaining lasers (the 633 nm laser always uses a 3-channel array).

Build	Installed lasers (nm)	Front 8-channel array	Rear 8-channel array
2-laser	488, 633	488	None
3-laser	488, 561, 633	561	None
3-laser	488, 445, 633	488 / 445	None
3-laser	488, 633, 405	488	405
3-laser	488, 633, 375	488	375
4-laser	488, 561, 633, 405	405	561
4-laser	488, 561, 633, 375	375	561
4-laser	488, 445, 561, 633	561	None
4-laser	488, 445, 633, 405	488 / 445	405
4-laser	488, 445, 633, 375	488 / 445	375
4-laser	488, 633, 405, 375	488	405 / 375
5-laser	488, 561, 633, 405, 375	405 / 375	561
5-laser	488, 445, 561, 633, 405	405	561
5-laser	488, 445, 561, 633, 375	375	561
5-laser	488, 445, 633, 405, 375	488 / 445	405 / 375
6-laser	488/445, 561, 633, 405/375	405 / 375	561

Reference optical maps

Mirror and filter placement for the standard 2-laser build and the full 6-laser build. The mirror at each position is a longpass dichroic; the filter is the bandpass in front of that PMT. The last active position in each array has no mirror.

Build	Array (laser)	PMT	LP mirror	BP filter	Typical dyes
2-laser	8-ch (488)	A / B / C	735 / 655 / 610	780/60 / 695/40 / 610/20	PE-Cy7 / PerCP-Cy5.5, PI / PE-Texas Red
		D / E / F	556 / 502 / —	585/42 / 530/30 / 488/10	PE / FITC / SSC
	3-ch (633)	A / B	735 / —	780/60 / 660/20	APC-Cy7 / APC
6-laser	3-ch (488)	A / B / C	655 / 502 / —	695/40 / 530/30 / 488/10	PerCP-Cy5.5 / FITC / SSC
	3-ch (445 alt.)	A / B	470 / —	510/80 / 445/15	CFP, AmCyan / SSC
	8-ch (561)	A / B / C / D	735 / 630 / 600 / —	780/60 / 670/14 / 610/20 / 582/15	PE-Cy7 / PE-Cy5 / PE-Texas Red, mCherry, PI / PE
	3-ch (633)	A / B	735 / —	780/60 / 660/20	APC-Cy7 / APC
	8-ch (405)	A / B	502 / —	510/50 / 450/40	AmCyan / DAPI, Pacific Blue
	8-ch (375 SP)	A / B	610 / —	670 LP / 450/20	Hoechst Red / Hoechst Blue (side population)

Source: CFG-CSI-001 §3 and §4; DWG-CSI-5100; DWG-CSI-5200. Configuration rules to enforce: (1) when the 445 nm laser is active, the 405 nm array bandpass changes from 450/40 to 480/20; (2) the 3-PMT 633 nm upgrade replaces the 735 LP with a 755 LP and adds a 730/45 BP; (3) the 5-PMT 561 nm option adds a 685 LP and 710/50 BP and shifts downstream optics. Each rule is a kitting check and an FAT check.

3.8 Fluidics

The fluidics supply module is a self-contained mobile unit. It generates its own air pressure and vacuum and supplies filtered sheath at constant pressure. Sheath pressure is set by the software according to the sort setup chosen for the installed nozzle. Changing sheath pressure changes drop frequency, drop delay, laser delay and area scaling, which is why the factory stores a verified setup for each nozzle.

Fluidics parameter	Specification	Source document
Sheath pressure range	5–75 PSI, software-controlled	SSD-CSI-100 §4.1
Module air supply	On-board compressors; supply gauge 80–100 PSI with stream on ($\approx$ 88 PSI typical); out-of-range reading means a fault	SSD-CSI-100 §4.2
Sheath tank	10 L stainless steel, pressurized, autoclavable (125 °C, 15 psig, 30 min)	SSD-CSI-100 §5.1; DWG-CSI-1100
Ethanol shutdown tank	5 L stainless steel, 70% ethanol	SSD-CSI-100 §5.2
Waste container	10 L plastic, 6-level sensor probe, disposable vented cap	SSD-CSI-100 §5.3
Cleaning fluid containers	Three 5 L plastic: 10% bleach, DI water (with 3 mL bleach per L), 70% ethanol	SSD-CSI-100 §5.4
Filtration	Sheath filter (also used on ethanol tank), fluid filters on cleaning tanks, 3 μ m inline air filter, sheath tank air filter	SSD-CSI-100 §5.5
Level sensing	Sheath, waste and auxiliary sensor probes (ethanol-rated probe on the ethanol line)	SSD-CSI-100 §6
Sheath fluid	Buffered saline sheath or 1× PBS; low-detergent only, to prevent foaming	OG-CSI-001 §5.3
Sample flow rate	Software settings 1–11, about 10–80 μ L/min	SSD-CSI-100 §7.1
Sample chamber	Pressurized, motorized loading port; agitation at selectable speeds; temperature hold (about 45 min to reach setpoint)	SSD-CSI-100 §7.2

Fluidics parameter	Specification	Source document
Sample tube formats	15 mL conical, 12 × 75 mm, 1 mL microtube (dedicated holders)	SSD-CSI-100 §7.3; ACC-CSI-001
Sample line	Replaceable sample injection tubing (12 in and 7 in), optional 35 or 50 µm inline filter	SSD-CSI-100 §7.4
Carryover control	Automatic inside/outside flush of the sample line after every tube; backflush mode	SRS-CSI-001 §4.8
Stream stability	Sheath line isolated from vibration; module casters locked with leveling feet	SSD-CSI-100 §8; SPG-CSI-001 §4

3.9 Flow cell, nozzles and sort setups

Nozzles lock into a keyed position at the lower end of the cuvette, below the interrogation point, so changing a nozzle does not affect optical alignment. The sort block pivots to recenter the stream in the waste aspirator if the keyed stream position differs between nozzles. Each nozzle size has a factory default setup, listed below, which the software loads automatically; a nozzle must never be run with another size's setup.

Setting	70 µm	85 µm	100 µm	130 µm (opt.)
Sheath pressure (PSI)	70	45	20	10
Drop drive frequency (kHz)	87	47	30	12
Drop drive amplitude (V)	60	32	12	24
Drop 1 target (pixels)	150	150	150	150
Gap target (upper limit), pixels	6 (14)	7 (17)	10 (21)	12 (21)
Drop delay (drops)	47	30	27	16
Plate voltage (V)	4,500	4,000	2,500	2,000
Stream charge far L / L / R / far R (%)	100 / 40 / 40 / 100	100 / 35 / 35 / 100	80 / 30 / 30 / 80	60 / 20 / 20 / 60
2nd / 3rd / 4th drop charge correction (%)	20 / 10 / 0	20 / 10 / 0	10 / 5 / 0	0 / 0 / 0
Laser delay red / violet / yellow-green	−78 / 39 / −39	−71 / 35 / −35	−80 / 40 / −40	−78 / 39 / −39
Area scaling blue / red / violet / YG	0.70 / 0.60 / 0.50 / 0.80	1.00 / 0.75 / 0.50 / 0.80	0.90 / 0.65 / 0.50 / 0.90	0.90 / 0.65 / 0.50 / 0.95
Window extension	2.0	2.0	2.0	4.0

Source: SSD-CSI-700 §6 (default sort setup table); SSD-CSI-300 §4 (nozzles). Also supplied: a closed-loop nozzle for cleaning and shutdown. Nozzles use a replaceable seal (a standard O-ring is an approved temporary replacement). Values are typical defaults; per-unit values are recorded in the Device History Record (DHR).

3.10 Sort system

Sort parameter	Specification	Source document
Drop drive	Frequency 1.0–102.0 kHz; amplitude 1.0–80.0 V; attenuation available for low pressures	SSD-CSI-700 §3.1
Breakoff imaging	Upper camera with LED strobe imaging just below the nozzle; strobe phase adjustable 0–360°	SSD-CSI-700 §3.2
Automatic breakoff control	Closed loop on drop drive amplitude to hold Drop 1 (100–600 px; acceptable within 10 px of target) and Gap. If Gap drifts more than 1 px, amplitude is adjusted; more than 2 px pauses sorting until recovered. A clog stops the stream, turns off plate voltage, closes the aspirator drawer and unloads the tube	SSD-CSI-700 §3.3; SRS-SW-CSI-010 §6.4
Drop delay	10–140 drops; set with a 660 nm diode laser, side-stream camera and switchable emission filter using fluorescent drop-delay beads; automatic drop-delay routine in software	SSD-CSI-700 §3.4
Charging	Charging wire in the flow cell; up to 4 sort streams (far left, left, right, far right) plus center waste	SSD-CSI-700 §4.1
Deflection	Two HV plates, plate voltage up to 4,500 V setting; potential between plates up to 12,000 V; red warning light when energized; plates stay energized with the sort block door open	SSD-CSI-700 §4.2; DWG-CSI-7000
Event timing resolution	Particle position resolved to 1/32 of a drop	SSD-CSI-600 §6.1
Conflict masks	Yield, purity and phase masks, each 0–32 (in 1/32-drop steps); yield and phase are mutually exclusive	SRS-SW-CSI-010 §7.2
Precision modes	Purity (32/32/0), 4-Way Purity (0/32/0), Yield (32/0/0), Single Cell (0/32/16), Initial (32/0/0), Fine Tune (0/0/0) (yield / purity / phase)	SRS-SW-CSI-010 §7.3
Tube collection	2-way: 12 × 75 mm, 15 mL. 4-way: 12 × 75 mm, 1.5 mL, 1 mL microtube (universal top with aerosol vent holes)	SSD-CSI-800 §3
Plate and slide deposition (option)	Automated plate deposition unit: 6-, 24-, 48-, 96- and 384-well plates, microscope slides, micro-well trays via adapter; custom device definitions; index sorting records the parameters of each deposited cell	SSD-CSI-800 §4
Waste aspirator	Central aspirator with motorized drawer that closes to protect tubes on clog or e-stop	SSD-CSI-700 §5

3.11 Electronics, safety controls and software

Element	Description	Source document
Signal processing	Digital pulse processing (height, area, width) per detector; laser delay and area scaling per laser; card-cage architecture	SSD-CSI-600 §4
Acquisition rate	Up to 70,000 events/s with 8 parameters and 12 compensation pairs; > 100,000 events/s with fewer parameters	SRS-CSI-001 §4.6
Power panel	Main power (sorter and fluidics module), circuit breaker, per-laser rotary switches	SSD-CSI-600 §3
Emergency stop	Illuminated button next to the loading port. Unloads the tube, stops plate stage motion, turns off stream and plate voltage, closes the aspirator drawer and shows a warning. Does not turn off lasers or main power	SRS-CSI-001 §5.4; SSD-CSI-600 §7
Interlocks	Flow cell access door laser shutter; sort block door must be closed before the collection chamber door opens	SRS-CSI-001 §5.3
Workstation	Desktop PC (Windows), one or two monitors, printer; Ethernet link to instrument	ICD-CSI-020 §3
Instrument control software	Acquisition and analysis, automated fluidics startup, shutdown and cleaning, sort setup, stream monitoring, clog detection and recovery, bead-based performance QC with automatic laser delay, configuration management of mirrors and filters	SRS-SW-CSI-010 §3
IVD software additions	User access control, audit trail, locked IVD assay templates, electronic records	SRS-SW-CSI-010 §9

3.12 Performance specifications

Characteristic	Specification	Source document
Fluorescence resolution (PI)	CV < 3.0%, full G0/G1 peak, PI-stained chicken erythrocyte nuclei (CEN), 488 nm	SRS-CSI-001 §4.1; PVR-CSI-001
Fluorescence resolution (Hoechst)	CV < 3.5%, full G0/G1 peak, Hoechst-stained CEN, 405 or 375 nm	SRS-CSI-001 §4.1; PVR-CSI-001
Fluorescence linearity	Doublet-to-singlet ratio 1.95–2.05 (CEN with PI off 488 nm, or Hoechst off 405/375 nm)	SRS-CSI-001 §4.2; PVR-CSI-001
Scatter sensitivity	Resolves fixed platelets from noise, identifies bacteria and 0.5 µm beads	SRS-CSI-001 §4.3
Acquisition rate	70,000 events/s (8 parameters, 12 compensation pairs)	SRS-CSI-001 §4.6
Sort purity and yield	Purity > 98% and yield > 80% of Poisson expected, 4-way sort at 70 PSI, 90 kHz, 25,000 events/s average threshold rate	SRS-CSI-001 §4.7; PVR-CSI-001
Viability	Cell lines and human PBMCs sorted across sheath pressures remained viable and proliferated for several days after sort	PVR-CSI-001 §9

3.13 Options

Option	Components	Key specifications	Source document
Aerosol evacuation module	Mobile evacuator; ULPA filter module with tubing; air filter for collection chamber door; hydrophobic filter on sort block door; hinged sample chamber cover	≥ 7 CFM normal (20% suction), ≥ 30 CFM boost (100%); three-stage ULPA (pre, ULPA, post), > 99.9995% at ≥ 0.12 µm; < 35 lb	SSD-CSI-900 §3
Collection temperature control	Recirculating water bath; insulated hoses; jacketed 2-way 15 mL, 4-way 12 × 75 mm and 4-way 1.5 mL holders	Bath range –20 to 120 °C; ambient 5–40 °C; offset table (e.g. 37.5 °C bath for 37 °C sample); distilled water with 0.1 g/L sodium carbonate	SSD-CSI-950 §3
Automated plate deposition unit	Motorized X-Y stage, plate loader, cabinet air filters	Plates 6 to 384 wells and slides; stage stops on e-stop	SSD-CSI-800 §4
Laser and detector upgrades	Laser, fiber head, detector array and PMT kits	Field-installable per configuration rules in 3.7	CFG-CSI-001 §5

3.14 Safety characteristics manufacturing must preserve

Hazard	Design control	Production verification
Laser radiation (Class 3B embedded)	Enclosure, shielding, door shutter interlock, warning labels	Interlock function test; label placement check; enclosure closure check
High voltage (up to 12 kV between plates)	Warning light, software indicator, insulation, e-stop turns off voltage	Hipot; warning-light and e-stop function test
Biohazard aerosols	Closed doors during sort, clog response, aerosol evacuation option	Containment verification on every unit
Moving parts (loading port, plate stage)	E-stop, illuminated button, guarded motion	E-stop response test
Pressurized vessels	Rated stainless tanks, relief protection	Pressure and leak test; tank certification check
Electrical shock / fire	IEC 61010 design, circuit breaker	Ground bond, hipot, leakage current

3.15 Accessory kit and ship-with items

CSI part no.	Item	Qty
CSI-ACC-101	Sample tube holders: 15 mL, 12 × 75 mm, 1 mL microtube	1 each
CSI-ACC-110	Collection holders: 2-way 15 mL, 2-way 12 × 75 mm, 4-way 12 × 75 mm, 4-way 1.5 mL; universal top	1 each
CSI-ACC-120	Keyed nozzles 70, 85, 100 µm and closed-loop cleaning nozzle (130 µm optional)	1 each
CSI-ACC-121	Nozzle locking lever assembly (spring and plunger), nozzle O-rings	1 set
CSI-ACC-130	Sample injection tubing, 12 in and 7 in; ferrules, collet, collet nut, ferrule removal tool	1 set
CSI-ACC-131	Pinch valve tubing; sample line filters 35 µm and 50 µm	1 set
CSI-ACC-140	Sheath filter; cleaning-fluid filters; 3 µm inline air filter	1 set
CSI-ACC-141	Disposable waste caps; 5 L container caps; O-ring lubricant; magnifier	1 set
CSI-ACC-150	FSC ND filters 0.5, 1.5, 2.0 (1.0 installed); spare filter and mirror holders	1 set
CSI-ACC-160	Tools: O-ring pick, deflection plate removal tool, Allen set, sheath probe wrench	1 set
CSI-ACC-170	Regional power cords (as ordered)	1
CSI-QC-200	Performance QC beads, drop-delay beads, alignment beads (ship-with starter lot)	1 lot

3.16 Maintenance schedule (inputs to service and to life testing)

Maintenance task	Frequency	Manufacturing relevance
Purge fluid filters and sheath filter; check drip tray	Weekly	Drip tray and purge ports accessible and leak-free at build
Replace disposable waste cap	Monthly	Cap included in ship kit
Replace sheath filter	Every 3 months	Filter installed and primed at factory; lot recorded
Replace fluid filters, ethanol filter, sheath tank air filter	Every 6 months	Spares kitting
Replace sample lines	Every 4–6 months	Routing and ferrule seating per work instruction
Replace cabinet air filters	Every 6–12 months	Filter orientation check
Clean flow cell (DI water)	Daily shutdown or when CVs rise	Flow cell cleanliness at ship verified by CV test
Unscheduled: nozzle, pinch valve tubing, camera windows, deflection plates, optical filters, chamber O-ring	As needed	Service procedures validated on pilot units

4. Transfer Organization and Responsibilities

A cross-functional Design Transfer Team (DTT) is chartered at the start of design verification. Each subsystem has a named R&D owner and a named manufacturing receiver. A subsystem is not transferred until the receiver signs that they can build, test and troubleshoot it without R&D support.

Activity	Sys Eng	R&D	Mfg Eng	Quality	Supply Chain	RA	Service
Transfer plan & schedule	A/R	C	C	C	C	I	I
Product technical description	A/R	R	C	C	I	C	C
DMR content (drawings, specs, BOM)	C	R	C	A	C	I	I
Configuration definitions	A/R	R	C	C	C	I	C
Factory alignment & test specs	C	R	R	A	I	I	C
Work instructions & routings	C	C	A/R	C	I	I	C
Process FMEA	C	C	R	A	C	I	I
Process validation (IQ/OQ/PQ)	C	C	R	A	I	I	I
Test method validation / transfer	C	R	R	A	I	I	C
Tooling & fixture qualification	I	C	A/R	C	C	I	I
Supplier qualification & AVL	I	C	C	A	R	I	I
Training & operator qualification	C	R	A	C	I	I	R
Labeling & IFU release	C	C	C	C	I	A/R	C
Transfer readiness reviews	A/R	R	R	R	R	C	C

R = Responsible, A = Accountable, C = Consulted, I = Informed

5. Transfer Strategy and Phases

Transfer is incremental rather than a single event at the end of development. Manufacturing engineers build prototype units during verification, so production documentation is exercised on real hardware long before the validation lots. Three gated reviews control progression.

Phase	Timing	Key activities	Exit gate
T0 Early engagement	During design verification	DFM/DFA reviews; draft pFMEA; identify special processes; long-lead laser and optics sourcing; Mfg Eng builds engineering units using draft instructions	Transfer plan approved
T1 Pilot build	After DV complete	Pilot lot of 5 units covering the 2-, 4- and 6-laser builds, built by production staff with redlined instructions; fixture IQ/OQ; test method validation; factory limits in FTS-CSI-001 drafted from pilot data	DTRR-1: documentation and equipment ready
T2 Validation build	Before design validation	PQ lots built to the released DMR; units feed design validation and regulatory testing; operator qualification; shipping validation	DTRR-2: process validated, DMR released
T3 Controlled ramp	First production months	Production under heightened monitoring; R&D on-call support; yield and nonconformance tracking	DTRR-3: transfer closure, handoff signed

From DTRR-1 onward, any design change touching a transferred subsystem requires an Engineering Change Order with a documented impact assessment on validated processes, test methods, factory limits and training. This protects validation evidence from being silently invalidated.

6. Device Master Record (DMR) Readiness

The DMR must be complete, released under document control, and sufficient for a trained operator to build, test, package and release any CSI-100 configuration without reference to R&D notebooks or verbal instruction. Readiness is assessed element by element against the checklist below.

DMR element	Content for CSI-100	Owner	Readiness evidence
Device specifications	DWG-CSI-0001 and subsystem drawings, BOM-CSI-0001, SCH-CSI-6000, PCBA fabrication packages, software/firmware bill of materials	R&D	All items at Released status in PLM; no open redlines
Configuration definitions	CFG-CSI-001: top-level assembly per build in 3.7, mirror and filter maps, software configuration files, upgrade kit BOMs and procedures	Sys Eng	Every released build assembled and passed FAT at least once
Factory specifications	FTS-CSI-001 (Section 7), with per-nozzle default setup files from SSD-CSI-700 §6	R&D / Quality	Limits justified by pilot capability data
Production process specs	Routings; WI-CSI-1001 to -1012 for subsystem build, laser integration, alignment, integration and burn-in; torque, adhesive and optical gel specs	Mfg Eng	Instructions executed on pilot lot with no uncontrolled deviations
Special processes	Fiber launch alignment, optical gel coupling, optics bonding, fluidic leak test, HV insulation, ESD handling	Mfg Eng	Validated per Section 8
Quality assurance procedures	INS-CSI-001 incoming inspection plans, in-process tests, FAT, sampling plans, DHR review	Quality	Test methods validated per Section 9
Software installation	Controlled software images per SRS-SW-CSI-010, install procedure, configuration and license activation, checksum verification	Software	Install verified on validation lot
Packaging & labeling	PKG-CSI-001 crating for sorter, fluidics module and options; LBL-CSI-001 labels; OG-CSI-001 IFU; UDI	RA / Mfg Eng	ISTA test passed; label review signed
Installation & service	SPG-CSI-001 (bench, 20 A outlet, clearances, optional house air); SM-CSI-001 installation qualification, field alignment check and maintenance procedures	Service	Executed by field engineers on pilot units

Traceability requirement

Every characteristic classified as critical in the design (laser power at each beam spot, beam spot geometry, inter-laser delay, CEN CVs, breakoff stability, drop-delay accuracy, sort purity, plate voltage, containment performance) must trace from a design output in the Design History File (DHF) to a specific DMR inspection or test step. The trace matrix (TM-CSI-DT-01 Rev B) is reviewed at each readiness gate; any critical characteristic without a production control blocks transfer.

7. Factory Build, Alignment and Test Specifications

This section summarizes the Factory Alignment and Test Specification (FTS-CSI-001 Rev A). The product specification column comes from the System Requirements Specification or the subsystem design specification cited; it is the release floor. Factory limits are tighter to leave margin for shipping, installation and field drift, and are finalized from pilot build capability data (target Cpk $\geq$ 1.33).

7.1 Build sequence

Step	Work instruction	Operation	Location	Key controls
1	WI-CSI-1001	Fluidics supply module build and pressure/leak test	Contract mfr	Pressure decay, level sensor calibration, FAI
2	WI-CSI-1002	Flow cell and objective coupling	ISO 7 cleanroom	Void-free gel coupling, cuvette face inspection
3	WI-CSI-1003	Excitation launch plate: fiber heads, prisms, focusing lens	Optics cell	Beam spot position and size on target
4	WI-CSI-1004	Laser integration per configuration	Optics cell	Laser power at head and at flow cell; fiber coupling efficiency
5	WI-CSI-1005	Detector arrays: PMTs, mirrors, filters per configuration map	ESD area	Kitting scan against configuration BOM
6	WI-CSI-1006	Collection fiber alignment to beam spots	Optics cell	Bead signal maximized; fiber position locked
7	WI-CSI-1007	Sort block, HV plates, cameras, drop-delay diode	Final assy	Hipot; camera focus and calibration
8	WI-CSI-1008	Electronics, power panel, interlocks, e-stop	Final assy	Functional and safety tests
9	WI-CSI-1009	Software install, configuration file, per-nozzle setups	Test bay	Checksum and configuration readback
10	WI-CSI-1010	System alignment, QC baseline, sort performance tests	Test bay	Section 7.3 FAT
11	WI-CSI-1011	Burn-in (48 h stream and laser cycling) and post-burn-in retest	Burn-in racks	Drift within FTS-CSI-001 limits
12	WI-CSI-1012	Pack, crate, ship (sorter, fluidics module, options)	Shipping	Shock and tilt indicators; lasers locked out

7.2 Alignment specifications

Parameter	Product spec	Factory limit	Source document
Laser power out of head (incoming)	Per 3.5 minimums	$\geq$ minimum + 10%	SSD-CSI-400 §3; FTS-CSI-001 §3.1
Beam spot height $\times$ width	$9 \pm 3 \mu\text{m} \times 65 \pm 7 \mu\text{m}$	$9 \pm 2 \mu\text{m} \times 65 \pm 5 \mu\text{m}$	SSD-CSI-400 §4.1; FTS-CSI-001 §3.2
Beam spot vertical position on core stream	Centered on core	Peak bead signal within $\pm 2 \mu\text{m}$ of spot center	FTS-CSI-001 §3.3
Inter-laser delay (70 μm setup)	Defaults per 3.9	Within ± 3 units of default	SSD-CSI-600 §5.3; FTS-CSI-001 §3.4
Alignment bead rCV, primary channel per laser	—	$\leq 3.0\%$ (FSC, SSC and primary PMT)	FTS-CSI-001 §3.5
Collection fiber alignment	Fixed 200 μm apertures	Bead signal $\geq 95\%$ of peak at locked position	SSD-CSI-500 §5; FTS-CSI-001 §3.6
Stream centering in waste aspirator	Stream enters aspirator	Center within $\pm 0.25 \text{ mm}$, all four nozzle sizes	SSD-CSI-300 §4.3; FTS-CSI-001 §3.7
Nozzle-change repeatability	Alignment unaffected	Stream position change $\leq 0.1 \text{ mm}$ across 10 reinsertions	SSD-CSI-300 §4.4; FTS-CSI-001 §3.8
Breakoff camera focus and scale	Drop 1 range 100–600 px	Drop spacing $\approx 57 \text{ px}$ at 70 μm setup, $\pm 2 \text{ px}$	SSD-CSI-700 §3.2; FTS-CSI-001 §3.9
Side-stream deflection symmetry	—	Left and right streams symmetric within $\pm 0.5 \text{ mm}$ at collection plane	FTS-CSI-001 §3.10

7.3 Final acceptance test (FAT)

Every unit passes the full FAT in its shipped configuration. Tests with a product specification use a guard-banded factory limit.

Test	Method	Product spec	Factory limit	Source document
Fluorescence resolution (PI)	CEN, 488 nm, full G0/G1	CV < 3.0%	CV $\leq 2.7\%$	SRS-CSI-001 §4.1; FTS-CSI-001 §5.1
Fluorescence resolution (Hoechst)	CEN, 405 or 375 nm	CV < 3.5%	CV $\leq 3.2\%$	SRS-CSI-001 §4.1; FTS-CSI-001 §5.2
Linearity	CEN doublet / singlet	1.95–2.05	1.97–2.03	SRS-CSI-001 §4.2; FTS-CSI-001 §5.3
Scatter sensitivity	0.5 μm beads vs. noise	Resolved	Resolved with ND 1.0 removed	SRS-CSI-001 §4.3; FTS-CSI-001 §5.4
Acquisition rate	8 parameters, 12 comp. pairs	70,000 events/s	No data loss at 75,000 events/s	SRS-CSI-001 §4.6; FTS-CSI-001 §5.5
Performance QC baseline	QC bead run, all detectors	Pass	Pass; baseline stored in DHR	SRS-SW-CSI-010 §8; FTS-CSI-001 §5.6
Breakoff stability	70 μm , breakoff control on, 4 h	Within 10 px of Drop 1 target	$\leq 5 \text{ px}$ drift; Gap within target +1	SSD-CSI-700 §3.3; FTS-CSI-001 §5.7
Drop delay	Automatic routine with drop-delay beads	Converges	Within ± 0.25 drop of manual method	SSD-CSI-700 §3.4; FTS-CSI-001 §5.8
Sort purity / yield	4-way bead sort, 70 PSI, 90 kHz, 25,000 events/s	> 98% / > 80% Poisson	$\geq 99\%$ / $\geq 85\%$	SRS-CSI-001 §4.7; FTS-CSI-001 §5.9

Test	Method	Product spec	Factory limit	Source document
Plate deposition (option)	Single-bead sort into 96- and 384-well plates	—	≥ 98% of wells with exactly one bead	SSD-CSI-800 §4; FTS-CSI-001 §5.10
Clog response	Simulated clog	Stream off, HV off, drawer closed, tube unloaded	All four actions within 1 s	SRS-CSI-001 §5.5; FTS-CSI-001 §5.11
Containment (with aerosol option)	Fluorescent microsphere challenge with simulated clog	—	No particles detected outside the sort chamber	SRS-CSI-001 §5.6; FTS-CSI-001 §5.12
Evacuator flow	Flow gauge / anemometer	≥ 7 CFM / ≥ 30 CFM	≥ 7.5 / ≥ 32 CFM	SSD-CSI-900 §3.2; FTS-CSI-001 §5.13
Sample chamber temperature	Setpoint hold	Reaches setpoint in ≈ 45 min	±1 °C at 45 min	SSD-CSI-100 §7.2; FTS-CSI-001 §5.14
Fluidics pressure	Supply gauge with stream on	80–100 PSI	84–96 PSI	SSD-CSI-100 §4.2; FTS-CSI-001 §5.15
Electrical safety	Ground bond, hipot, leakage	IEC 61010-1	100% pass	SRS-CSI-001 §5.1; FTS-CSI-001 §5.16
Acoustic noise	Sound level at operator position	≤ 62 dBA	≤ 60 dBA	SRS-CSI-001 §3.4; FTS-CSI-001 §5.17
E-stop and interlocks	Functional test	Per 3.11	100% pass	SRS-CSI-001 §5.4; FTS-CSI-001 §5.18

8. Manufacturing Process Validation

Each production process is classified per the Master Validation Plan. Processes whose output cannot be fully verified by later inspection or test are validated through installation, operational and performance qualification (IQ, OQ and PQ). Processes that are fully verified are controlled through validated test methods instead.

Process	Subsystem	Verify or validate	Rationale
Fiber launch and beam-spot alignment	SS-04	Validate	Alignment retention after shipping and thermal cycling cannot be fully verified at the station
Flow cell to objective optical gel coupling	SS-03	Validate	Voids or contamination are latent; coupling can degrade over time
Collection fiber and detector array assembly	SS-05	Validate	Fiber position and adhesive cure cannot be fully inspected after assembly
Fluidics module build and leak test	SS-01	Validate	Long-term seal integrity of pressurized tanks and lines; leak test requires method validation
Nozzle keying and stream alignment	SS-03	Verify	Fully characterized by stream position and nozzle-change test
HV deflection assembly and insulation	SS-07	Validate	Insulation integrity; 100% hipot plus validated assembly
Aerosol chamber sealing	SS-09	Validate	Containment relies on seals that cannot be seen after closure
PCBA build	SS-06	Validate at vendor	Solder joints; covered by vendor process validation
Software install and configuration load	SS-10	Verify	Checksum, version and configuration readback on every unit
Final integration and FAT	All	Verify	Full functional test with validated methods

Stage	Objective	Typical evidence
IQ	Equipment, fixtures and software tools are installed and configured to specification	Calibration records, utilities, fixture drawings verified, software version records
OQ	Process produces acceptable output across the full range of operating parameters	Worst-case runs at upper and lower process limits (gel volume, cure time, torque); DOE where parameters interact
PQ	Process consistently produces acceptable output under normal production conditions	At least three consecutive lots, multiple operators and shifts, statistically justified sample sizes

Sample sizes are set by the pFMEA risk level of the characteristic (for example, 95% confidence / 95% reliability for high-risk attributes). Capability targets are $C_{pk} \geq 1.33$ for variable characteristics, and $C_{pk} \geq 1.67$ for characteristics linked to hazards of severity 4 or higher.

9. Test Method and Calibration Transfer

Instrument performance is proven in production by tests, so the tests themselves must be validated before their results can be trusted. Each production test method moves from R&D with a method validation report showing that it reliably separates good units from bad.

Test method	Measures	Validation approach
Optical alignment verification	Signal rCVs, relative intensity and inter-laser delay using alignment beads per laser	Gauge R&R across operators and fixtures; % study variation $\leq 10\%$ ($\leq 30\%$ conditional)
CEN resolution and linearity	G0/G1 CV and doublet/singlet ratio	Reference-lot characterization; comparison with golden instrument
Stream and breakoff stability	Stream position, breakoff drift, nozzle-change repeatability	Repeatability study; camera scale verified with calibration target
Drop delay and sort purity/yield	Automatic drop delay; purity and yield with mixed bead populations in tubes and plates	Comparison against reference re-analysis; seeded-fault units (wrong drop delay, weak charge)
Containment verification	Aerosol escape under simulated clog	Validated against an independent reference particle measurement
Electrical safety	Hipot, ground bond, leakage	Calibrated safety tester; test program under software control
Final acceptance test	Integrated performance against Section 7.3	Each limit traced to SRS requirement; guard band from measurement uncertainty

Reference standards and golden units

- Two golden reference instruments (6-laser build) are characterized and held under quality control: one stays with R&D and one moves to manufacturing for daily correlation.
- Chicken erythrocyte nuclei, alignment beads, performance QC beads and drop-delay beads are specified in the DMR, with lot qualification criteria.
- Test software used for automated production testing is validated per the software tool validation procedure.

10. Manufacturing Equipment, Tooling and Facilities

Equipment / fixture	Used for	Qualification
ISO 7 cleanroom bench with laminar flow hood	Flow cell, objective and nozzle handling	Particle counts per ISO 14644-1
Optical table with vibration isolation	Launch plate and fiber alignment	Flatness and isolation check
Beam profiler camera with flow-cell target fixture	Beam spot size and position	IQ/OQ; calibrated reference reticle
Optical power meter with fiber and free-space heads	Laser power at head and at flow cell	Annual calibration
Gel dispense and cure fixture	Optical gel coupling	OQ at gel volume and cure extremes
Spectrophotometer (filter verification)	Incoming mirror and filter spectra	Calibration with NIST-traceable standards
Nozzle keying gauge and stream position target	Nozzle and aspirator alignment	Gauge R&R
Camera calibration target set	Breakoff and side-stream camera scale and focus	Dimensional certification
Pressure decay leak tester	Fluidics module and tank integrity	Method validation
Electrical safety tester	Hipot, ground bond, leakage	Annual calibration
Aerosol challenge rig	Containment verification	Method validation
Anemometer / flow meter	Evacuator flow	Annual calibration
Burn-in racks with fluidics supply	48 h stream and laser cycling	IQ; monitoring software validated
Golden instruments (2)	Correlation and test-method reference	Characterization report
ESD workstations and PMT handling kits	Detector and PCBA handling	ANSI/ESD S20.20 audit
Laser safety controlled area	All open-beam work	Laser safety officer approval; eyewear by wavelength

11. Supplier Readiness and Approved Vendor List

Several performance-critical components come from specialist vendors. Supplier readiness is part of transfer, not a procurement afterthought. Vendor names are coded; the Approved Vendor List (AVL-CSI-001 Rev B) holds the identities, and incoming inspection follows INS-CSI-001.

Component	Vendor	Sourcing	Readiness requirement
488 / 445 / 405 / 561 nm solid-state lasers, fiber-coupled	Vendor L1 (L2 qualifying)	Dual (planned)	Power and beam-quality incoming test; change-notification agreement; wavelength-specific FAI
633 nm helium-neon laser	Vendor L3	Single	Tube life data; incoming power and warm-up stability; last-time-buy plan
375 nm solid-state laser (fixed mount)	Vendor L4	Single	Mount interface FAI; power ≥ 7 mW top-hat profile
660 nm diode (drop-delay system)	Vendor L5	Dual	Incoming power check
Quartz cuvette flow cell	Vendor O1	Single	Optical face inspection, cleanliness spec, vendor process validation
Fluorescence objective (NA 1.2)	Vendor O2	Single	NA and imaging quality certificate; FAI
Dichroic mirrors and bandpass filters	Vendor O3, O4	Dual	Spectral certificate per lot; incoming spectral sampling
Optical fibers and fiber heads	Vendor O5	Single	Coupling efficiency test; connector geometry
Photomultiplier tubes	Vendor D1	Single	Gain and dark-current data per serial; ESD handling
FSC photodiode assembly	Vendor D2	Dual	Responsivity per lot
Fluidics supply module (build to print)	Vendor F1	Single	Vendor IQ/OQ/PQ approved; FAI; pressure-vessel certification; shipping validation
Stainless sheath and ethanol tanks	Vendor F2	Dual	Pressure rating certificate; autoclave cycle qualification
Keyed nozzles	Vendor F3	Single	Orifice diameter and keying inspection; lot traceability
PCBAs (signal processing, control)	Vendor E1	Single	Vendor process validation; ICT and functional test coverage report
High-voltage supply	Vendor E2	Single	Safety certification; 100% incoming functional check
Breakoff and side-stream cameras	Vendor E3	Dual	Resolution and frame-sync verification
ULPA filter module and evacuator	Vendor A1	Single	Certified efficiency per lot; flow verification
Recirculating water bath	Vendor A2	Catalog item	Temperature accuracy certificate
Workstation PC and monitors	Vendor W1	Catalog item	Locked image; model change control

All critical vendors must have a signed quality agreement, a completed first article inspection and at least one lot accepted through incoming inspection before DTRR-2. Single-source critical parts need a documented continuity plan (safety stock, last-time buy or second-source schedule).

12. Training and Qualification of Personnel

Training is structured by role and subsystem. Operators are qualified, not just trained: each must demonstrate the task on production hardware, with the result checked against acceptance criteria.

Role	Training document	Training content	Qualification method
Assembly technician	TRN-CSI-001; WI-CSI-1001 to -1012	Subsystem work instructions, ESD and cleanroom practice, gel and adhesive handling	Supervised build of a qualification unit; first-pass yield at target

Role	Training document	Training content	Qualification method
Optical alignment specialist	TRN-CSI-002; WI-CSI-1003, -1004, -1006	Fiber launch, beam-spot and collection-fiber alignment; configuration maps; laser safety (Class 3B)	Independent alignment of two units (different builds) meeting Section 7.2; laser safety certification
Test technician	TRN-CSI-001 §6; FTS-CSI-001	FAT procedures, CEN and bead preparation, sort tests, test software, nonconformance recording	Participation in Gauge R&R; full FAT under observation
HV and electrical safety	TRN-CSI-001 §8; WI-CSI-1007, -1008	HV assembly, interlock and e-stop verification	Written assessment plus practical
Quality inspector	TRN-CSI-004; INS-CSI-001	Incoming inspection (optics, lasers, PMTs), final inspection, DHR review	Review of pilot-lot DHRs with QA sign-off
Field service engineer	TRN-CSI-003; SM-CSI-001; SPG-CSI-001	Installation, site readiness, field alignment check, maintenance per 3.16, troubleshooting	Install and verify a pilot unit at a beta or internal site

Knowledge capture: during the pilot build, R&D engineers write a troubleshooting guide for each subsystem based on the failures actually encountered (stream deflection, unstable breakoff, high CVs, clogs, communication faults). These become controlled DMR documents rather than tribal knowledge.

13. Risk Management in Transfer

The process FMEA is built from the design FMEA and the ISO 14971 hazard analysis, so every design risk control that depends on correct manufacture has a matching process control.

Transfer risk	Potential effect	Control in production
Beam-spot or fiber misalignment during shipping	Degraded sensitivity; misclassified cells; wrong sort	Validated crating; alignment check at installation qualification
Gel coupling void or contamination	Loss of collection efficiency; high CVs	Validated dispense and cure; cleanroom; CEN CV test
Wrong mirror or filter installed for configuration	Wrong channel assignment; erroneous results	Kitting scan to configuration BOM; FAT spectral check with single-stain beads
Configuration file does not match hardware	Detector map wrong in software	Configuration readback compared with build record
Drop-delay drift not detected	Reduced purity of sorted population	Drop delay verified in FAT; daily QC in software
Containment seal mis-assembly	Operator exposure to aerosols from infectious samples	Validated assembly; containment verification on every unit; interlock test
HV insulation defect	Electric shock hazard	Validated HV assembly; 100% hipot; warning-light test
Nozzle keying variation	Stream misses aspirator after nozzle change	Keying inspection; nozzle-change repeatability in FAT
Helium-neon laser end of life or supply loss	Line stoppage; field failures	Incoming life data; continuity plan; service stock
Undocumented R&D rework	Units built that cannot be reproduced	Any pilot deviation must result in a DMR update before DTRR-2

14. Transfer Acceptance Criteria

Transfer is complete only when every criterion below is met with objective evidence, reviewed at DTRR-2 and confirmed at DTRR-3. Criteria are pass/fail; conditional acceptance requires a documented risk assessment and Quality approval with a closure date.

#	Criterion	Evidence	Gate
AC-1	DMR complete and released; no open redlines or Draft-status documents	DMR index; PLM report	DTRR-2
AC-2	All critical characteristics traced from design output to production control	Trace matrix TM-CSI-DT-01	DTRR-2
AC-3	Every released configuration built and passed FAT at least once	Configuration build log; DHRs	DTRR-2
AC-4	Factory limits in FTS-CSI-001 justified by pilot capability data	FTS-CSI-001 justification report	DTRR-2
AC-5	All processes classified 'validate' have approved IQ/OQ/PQ reports	Validation summary report	DTRR-2
AC-6	All production test methods validated; Gauge R&R within limits	Method validation reports	DTRR-2
AC-7	Three consecutive validation units pass FAT first time with no DMR deviations	DHRs; FAT records	DTRR-2
AC-8	Tooling and fixtures qualified; golden units characterized	IQ/OQ records; characterization report	DTRR-2

#	Criterion	Evidence	Gate
AC-9	Critical vendors qualified, with quality agreements, accepted first articles and continuity plans	AVL; FAI reports	DTRR-2
AC-10	All production and field roles qualified	Training matrix; qualification records	DTRR-2
AC-11	pFMEA complete; no residual high risks without approved justification	pFMEA; risk report update	DTRR-2
AC-12	Shipping validation and installation qualification passed for sorter and fluidics module	ISTA report; IQ records	DTRR-2
AC-13	First-pass yield $\geq$ target and no open critical nonconformances over ramp period	Yield and NCR trend data	DTRR-3
AC-14	Manufacturing troubleshoots production failures without R&D intervention	Escalation log	DTRR-3

15. Post-Transfer Monitoring and Handoff Closure

During the controlled ramp (T3) the product stays under heightened monitoring with a defined exit, so that R&D support tapers off deliberately instead of stopping on a date.

- Weekly review of first-pass yield, nonconformances by subsystem and configuration, and FAT failure Pareto.
- Trending of installation failures, alignment drift and complaints from the first field units, fed back into the pFMEA.
- Laser and PMT field failure rates tracked by vendor and lot.
- R&D on-call support with a named contact per subsystem; each escalation logged and reviewed for DMR gaps.
- Lessons-learned review at DTRR-3, recorded in the DHF.
- Formal handoff: manufacturing and quality sign transfer closure; DMR ownership moves to sustaining engineering.

Design transfer is closed when the receiving organization can build, test, release, install and service every CSI-100 configuration consistently, using only controlled documentation, and has shown it on production volume, not just on validation units.

16. Reference Documents

The following controlled documents are referenced in this plan and together form the CSI-100 design and manufacturing baseline for transfer. The revisions listed are those released at DTRR-1. Later revisions apply once released through document change control. External standards are listed in Section 2.

16.1 Requirements and architecture

Document no.	Rev	Title	Owner
DHF-CSI-000	E	Design History File Index	Systems Eng
PRD-CSI-001	C	Product Requirements Document	Marketing
SRS-CSI-001	D	System Requirements Specification	Systems Eng
SRS-SW-CSI-010	E	Instrument Control Software Requirements Specification	Software
SAD-CSI-001	C	System Architecture Description	Systems Eng
ICD-CSI-010	B	Interface Control Document: Sorter to Fluidics Supply Module	Systems Eng
ICD-CSI-020	A	Interface Control Document: Sorter to Workstation	Systems Eng
CFG-CSI-001	C	Configuration Definition: Laser and Detector Builds	Systems Eng

16.2 Subsystem design specifications

Document no.	Rev	Title	Owner
SSD-CSI-100	C	Fluidics Subsystem Design Specification (supply module and sample delivery)	R&D Fluidics
SSD-CSI-300	B	Flow Cell and Nozzle Subsystem Design Specification	R&D Fluidics
SSD-CSI-400	C	Excitation Optics Subsystem Design Specification	R&D Optics
SSD-CSI-500	C	Collection and Detection Subsystem Design Specification	R&D Optics
SSD-CSI-600	B	Electronics and Signal Processing Subsystem Design Specification	R&D Electrical
SSD-CSI-700	C	Sort Subsystem Design Specification (includes default sort setup table)	R&D Sort
SSD-CSI-800	B	Sort Collection and Plate Deposition Subsystem Design Specification	R&D Mechanical
SSD-CSI-900	B	Aerosol Management Subsystem Design Specification	R&D Mechanical
SSD-CSI-950	A	Collection Temperature Control Option Design Specification	R&D Mechanical

16.3 Drawings, schematics and bills of materials

Document no.	Rev	Title	Owner
DWG-CSI-0001	C	Top-Level Assembly, CSI-100 Sorter (all configurations, tabulated)	R&D
DWG-CSI-1000	B	Fluidics Supply Module Assembly	R&D Fluidics
DWG-CSI-1100	B	Pressurized Sheath and Ethanol Tank Assembly	R&D Fluidics
DWG-CSI-3100	B	Flow Cell and Objective Assembly	R&D Optics
DWG-CSI-4000	C	Excitation Launch Plate Assembly	R&D Optics
DWG-CSI-5100	B	8-Channel Detector Array Assembly	R&D Optics
DWG-CSI-5200	B	3-Channel Detector Array Assembly	R&D Optics
DWG-CSI-7000	B	Sort Block and Deflection Plate Assembly	R&D Sort
SCH-CSI-6000	B	Electrical Schematics and Wiring Diagrams (set)	R&D Electrical
BOM-CSI-0001	C	Indented Bill of Materials, all configurations	R&D

16.4 Verification, safety and risk management

Document no.	Rev	Title	Owner
PVR-CSI-001	A	System Performance Verification Report	Systems Eng
SVR-CSI-001	A	Safety, EMC and Laser Compliance Test Report (IEC 61010, 61326, 60825)	Compliance
RMP-CSI-001	B	Risk Management Plan	Quality
RMR-CSI-001	A	Risk Management Report	Quality
HA-CSI-001	C	Hazard Analysis	Quality
DFMEA-CSI-001	C	Design Failure Mode and Effects Analysis	R&D
PFMEA-CSI-001	A	Process Failure Mode and Effects Analysis	Mfg Eng
TM-CSI-DT-01	B	Design Transfer Trace Matrix (design output to production control)	Systems Eng

16.5 Labeling, service and field documents

Document no.	Rev	Title	Owner
OG-CSI-001	B	Operator's Guide (Instructions for Use)	Technical Pubs
SM-CSI-001	A	Service Manual (installation, alignment check, maintenance, troubleshooting)	Service
SPG-CSI-001	A	Site Preparation Guide	Service
LBL-CSI-001	B	Labeling Specification (product, laser, HV and UDI labels)	Regulatory
ACC-CSI-001	B	Accessory Kit and Consumables Specification	R&D

16.6 Manufacturing and quality

Document no.	Rev	Title	Owner
MVP-CSI-001	B	Master Validation Plan	Quality
FTS-CSI-001	A	Factory Alignment and Test Specification	Mfg Eng / R&D
TMV-CSI-001	A	Test Method Validation Protocols and Reports (series)	Quality
INS-CSI-001	A	Incoming Inspection Plans: lasers, optics, PMTs, fluidics	Quality
PKG-CSI-001	A	Packaging and Shipping Specification	Mfg Eng
AVL-CSI-001	B	Approved Vendor List	Supply Chain

16.7 Manufacturing work instructions

Document no.	Rev	Title	Owner
WI-CSI-1001	A	Work Instruction: Fluidics supply module build and pressure/leak test	Mfg Eng
WI-CSI-1002	A	Work Instruction: Flow cell and objective coupling	Mfg Eng
WI-CSI-1003	A	Work Instruction: Excitation launch plate: fiber heads, prisms, focusing lens	Mfg Eng
WI-CSI-1004	A	Work Instruction: Laser integration per configuration	Mfg Eng
WI-CSI-1005	A	Work Instruction: Detector arrays: PMTs, mirrors, filters per configuration map	Mfg Eng
WI-CSI-1006	A	Work Instruction: Collection fiber alignment to beam spots	Mfg Eng
WI-CSI-1007	A	Work Instruction: Sort block, HV plates, cameras, drop-delay diode	Mfg Eng
WI-CSI-1008	A	Work Instruction: Electronics, power panel, interlocks, e-stop	Mfg Eng
WI-CSI-1009	A	Work Instruction: Software install, configuration file, per-nozzle setups	Mfg Eng
WI-CSI-1010	A	Work Instruction: System alignment, QC baseline, sort performance tests	Mfg Eng
WI-CSI-1011	A	Work Instruction: Burn-in (48 h stream and laser cycling) and post-burn-in retest	Mfg Eng
WI-CSI-1012	A	Work Instruction: Pack, crate, ship (sorter, fluidics module, options)	Mfg Eng

16.8 Training documents

Document no.	Rev	Title	Owner
TRN-CSI-001	A	Manufacturing Training Manual: Assembly and Test	Mfg Eng
TRN-CSI-002	A	Optical Alignment and Laser Safety Training Module	R&D Optics
TRN-CSI-003	A	Field Service Training Manual	Service
TRN-CSI-004	A	Quality Inspector Training: Incoming Inspection and DHR Review	Quality

Appendix A. Terms and Abbreviations

Term	Meaning
BP / LP / ND	Bandpass / longpass / neutral density optical filter
CEN	Chicken erythrocyte nuclei (resolution and linearity reference particle)
CV / rCV	Coefficient of variation / robust coefficient of variation
DHF / DMR / DHR	Design History File / Device Master Record / Device History Record
Drop 1 / Gap	Breakoff image metrics: position of the first free drop, and gap between stream and first drop
Drop delay	Time, in drops, from interrogation to breakoff; sets which drop is charged
DTRR	Design Transfer Readiness Review
FAT / FAI	Final Acceptance Test / First Article Inspection
FSC / SSC	Forward scatter / side scatter
IQ / OQ / PQ	Installation / Operational / Performance Qualification
PMT	Photomultiplier tube
pFMEA / dFMEA	Process / Design Failure Mode and Effects Analysis
ULPA	Ultra-low penetration air filter